



System for Detecting and Monitoring Respiratory System Related Diseases

Presented by **VitaSync**

Team Members

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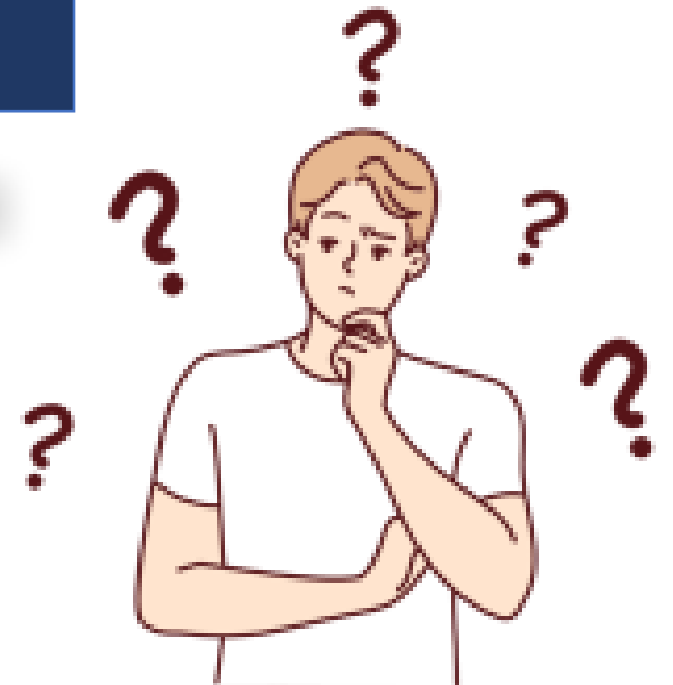
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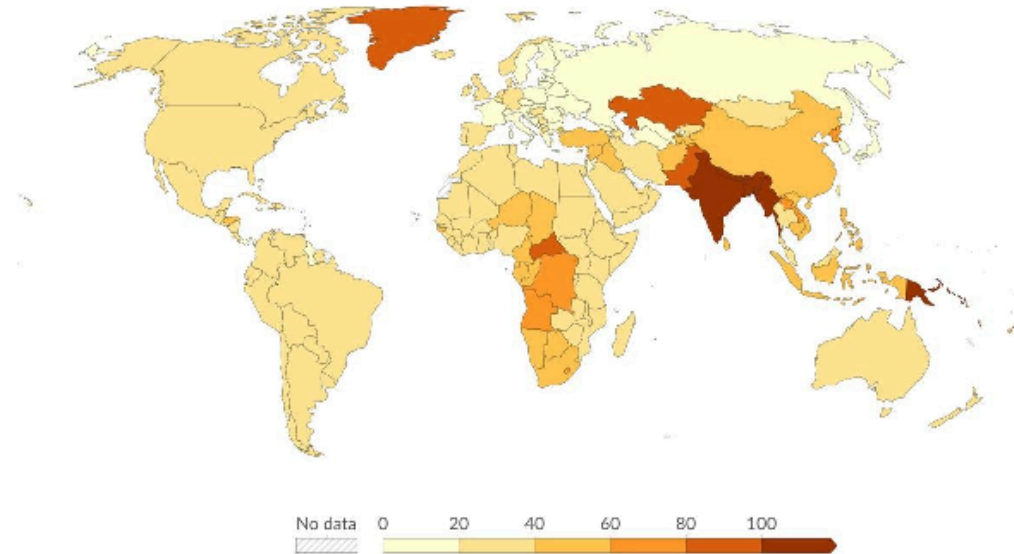
Problem Statement



Chronic respiratory diseases death rate, 2023

Estimated death rate from chronic respiratory disease¹ per 100,000 people.

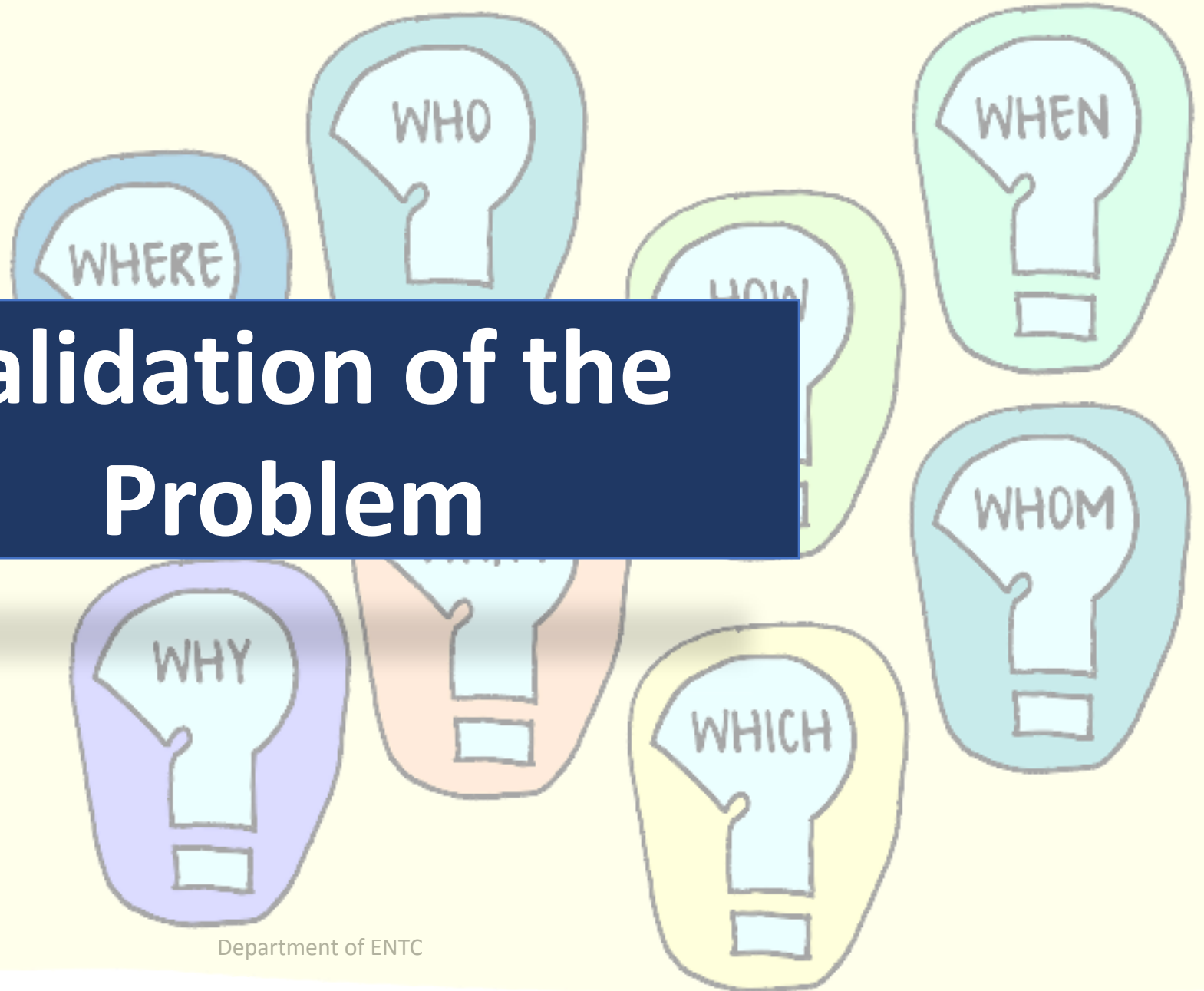
Our World
in Data



- **The Statistic:** Chronic respiratory diseases (CRDs) – primarily chronic obstructive pulmonary disease (COPD) and asthma – are a major and growing public health challenge in the WHO South-East Asia Region, responsible for nearly 12% of all deaths [WHO Link](#)
- **The Gap:** A critical lack of accessible, low-cost diagnostic and monitoring technology.
- **The Constraint:** Current tools are centralized in urban hospitals, leading to the neglect of early symptoms in rural populations



Validation of the Problem



Evidence: Hospital-grade spirometers are high-cost and inaccessible to the rural majority.

Technical Failure: Early signs are missed not just due to lack of awareness, but due to a lack of affordable,
point-of-care technology.

Engineering Demand: There is a clear need for a portable, non-invasive data logging solution.

NEWS FEATURE | 09 January 2026

India's silent lung crisis persists even as the world breathes easier

A global study shows deaths from chronic respiratory diseases are falling worldwide — but India and South Asia continue to carry heavy mortality burdens, driven by air pollution, smoking, and delayed diagnosis.

By [Subhra Priyadarshini](#)



Hospital grade spirometer that costs
around \$2000

Department of ENT

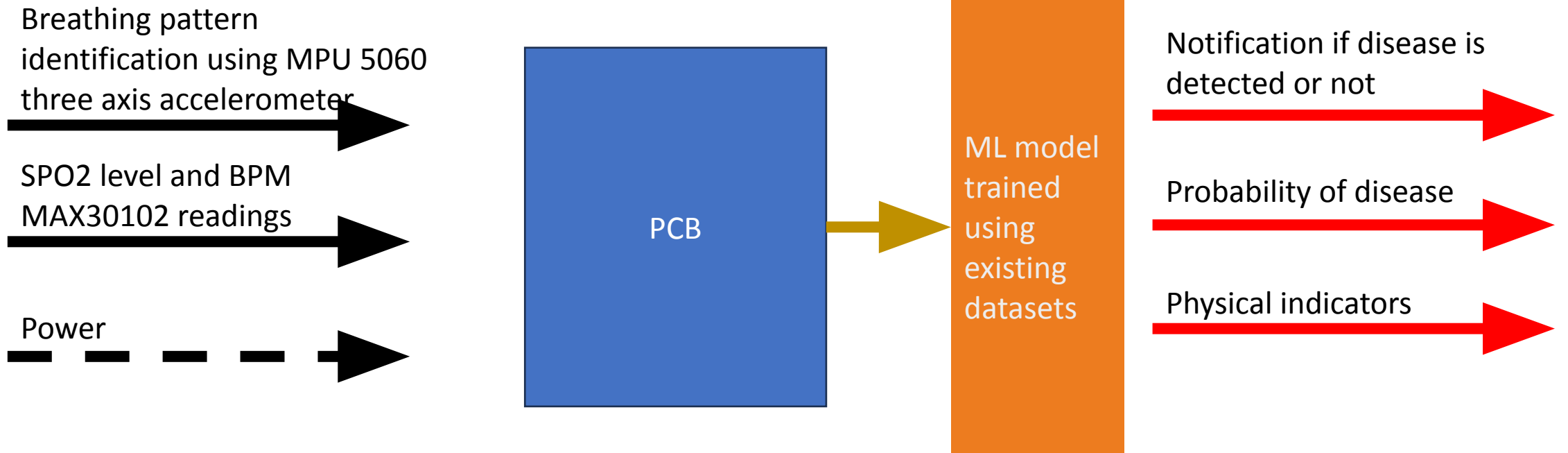


- Discussed the need for a non-invasive monitoring device with multiple resource personnel associated with the medical field.
 - Dr. Sanjeewa Tennekone - Consultant Neonatologist - 0773645577
 - Dr. Dilrukshi Maheshika - Consultant Rheumatologist - 0773417558
 - Dr. P.L.B.Nishantha - MO Cardiothoracic - 0776552891
 - Dr. Dammika Fernando - MO Radiology - 0777499150
 - Dr. S.A.P.L.Nanayakkara - 0772619993



Our Solution



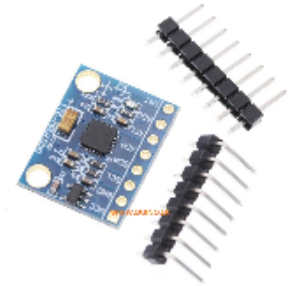


Hardware Used

Sensors



MAX30102



MPU 6050 3-Axis
Accelerometer

Micro-controllers



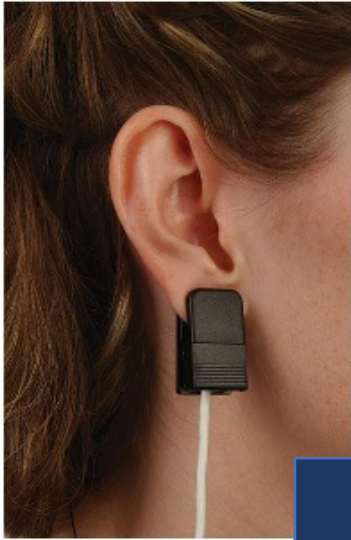
ATmega328P-AU



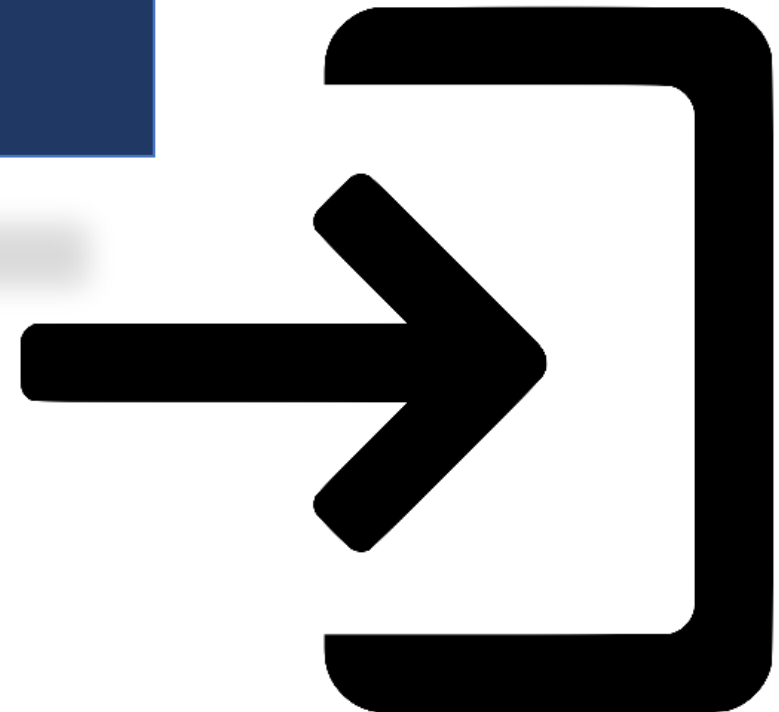
ESP32 D0WDQ6
(To run the ML model)

Other

Resistors
Capacitors
Voltage regulators
DC-DC Boost converters
Charging modules



Alternatives & Current Results



Alternatives Considered

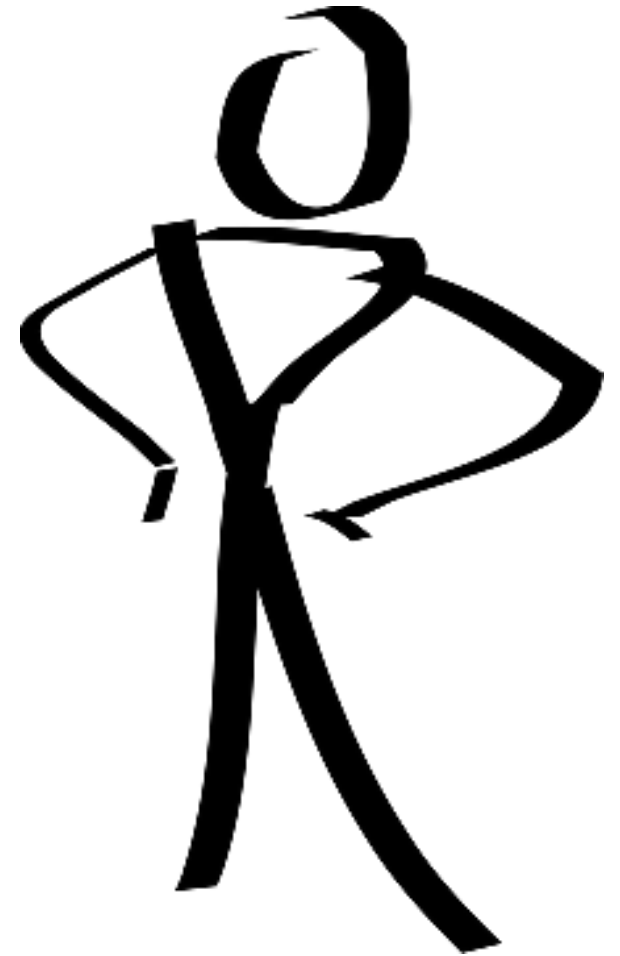
Measuring SpO2 levels through earlobe sensor.

Advantages : higher accuracy

Disadvantages : higher cost of implementation, difficulty in creating a comfortable wearable module.

Progress to Date

- **System architecture finalized** ✓
- **Sensor selection completed (PPG, Accelerometer).** ✓
- **Prototype implementation of the SPO2 sensor.** ✓
- **Calibration of SPO2 sensor** ✓
- **Schematic of SPO2 sensing module is completed** ✓



Challenges & Overcoming Them



Problem	Current solution
Amount of noise due to the distances from the sensor modules to the control unit and other noise related to the sensors are high.	Use a cable that can transmit sensor readings with less accumulation of noise (shielded 4-core wire). However, wearability issues might occur. Use digital filtering methods.
Creating enclosures for sensor modules that are comfortable to wear for users.	Use stretchable and elastic-like materials that are easy to wear.
Collecting data to train a ML model.	Use existing datasets that can be found on the internet.
Obtaining accurate readings from sensors.	Calibrate the sensors properly use digital filtering methods to remove noise.
Displaying the data and predictions to users.	Create a simple dashboard that can display readings and user information.

Thank you !